

A RESEARCH LOOP: UPDATE THE DESIGN

fixed LLM

Retained code, outcomes and failures guide the next round.

1 Hypothesis

testable change
expected effect

2 Source change

edit task code
feature / policy

4 Selection

score gate
keep / discard

5 Next round

source + scores
failed attempts

validate source

scores + diagnostics

Task contract: writable code | fixed trainer + scorer | roles + budgets

B FEDERATED EXECUTION

Proteomics / cells: 1 or 10 clients

Lab 1

D1

updates

Lab 2

D2

updates

Lab K

DK

updates

code + w

grad / w
count

3a Coordinator

aggregate
fit weights w

3b Evaluate

dev / CV
fixed scorer

C DOWNSTREAM TASKS

same interface; different adapters

Drug-target interaction

drug + protein $\rightarrow$ binding
central fit / frozen transfer
AUROC, AP

Proteomic efficacy

6h/24h proteomes + drug
federated logistic fitting
AP, AUROC, BCE

Perturbation ID (pilot)

single / double gene / drug
federated response heads
5-option Top-1, MRR